

Manufacturing Predictive Maintenance Blueprint

A practical enterprise decision guide from DigiScience Techsol

ASSET TYPE: PILOT PLANNING WORKSHEET

Illustrative planning method | Updated 20 September 2026

Use this worksheet to scope one plant workflow before proposing production rollout. It contains planning fields drawn from DigiScience's manufacturing service page, not measured customer outcomes, a validated prediction model or a safety approval.

Name the bounded decision

Record one target line or asset class, the maintenance signal to examine, the accountable business owner and one measurable KPI. Describe the existing action when a signal arrives. Keep plant experts accountable for operational decisions. The first proof may address one asset class, one inspection workflow or one maintenance signal. Do not combine all plant problems into a single uncontrolled pilot.

Assemble the input inventory

- Sensor history and alarms: identify the approved source and the target asset or line.
- Maintenance work orders and failure records: identify what maintenance action and failure information are available.
- Asset hierarchy and production context: document how the signal relates to the equipment and its operation.
- Quality events, operator notes and approved visual data: include these when relevant to the agreed workflow.
- Constraints: record the data access, integration path, operating restrictions and safety requirements that affect the pilot.

Map the signal to an action

Proposed reference flow: OT or IoT sources -> secure ingestion and storage -> feature or vision pipeline -> model and evaluation -> operator workflow -> CMMS or quality action -> monitoring and audit. For each stage, record the available source/system, responsible owner, expected output and unresolved dependency. The diagram is a planning reference, not an integration already deployed at a customer.

Choose measures before the pilot

Use the measures relevant to the selected problem: unplanned downtime, MTBF, maintenance lead time, false-alert rate, defect escape rate, review time, operator acceptance, and cost per prediction or inspection. Record the current baseline, proposed measurement source, accountable owner and agreed acceptance criterion. Leave unavailable values unknown. Do not insert a promised reduction, an invented baseline or a universal pass threshold.

Pilot evidence and controls

The pilot brief should identify the use-case baseline, approved sample data, a bounded prediction/classification/anomaly/vision/knowledge workflow, and governance controls. Plan human approval, audit trail, data classification, monitoring and failure-mode review. A scale recommendation should identify production architecture, integration backlog, cost view, rollout risks and success criteria. Feasibility evidence precedes production rollout; queue labels and dashboards do not establish that maintenance actions are staffed or effective.

Decision and source

Choose a bounded pilot only after reviewing data sources, integration, operating constraints, safety requirements and baseline measures. Scope implementation separately from a Solution Assessment; this PDF itself performs neither service. Source:
<https://digiscience.techsol.com/industries/manufacturing-ai/> Discuss one plant workflow:
<https://digiscience.techsol.com/contact>

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